

23CSE111 OBJECT ORIENTED PROGRAMMING

S2 B.Tech CSE

Department of Computer Science and Engineering
Amrita School of Computing, Amritapuri Campus

CASE STUDY

PHASE – 2 (Design)
Architectural Modelling (CO2)

Group No : **C4**

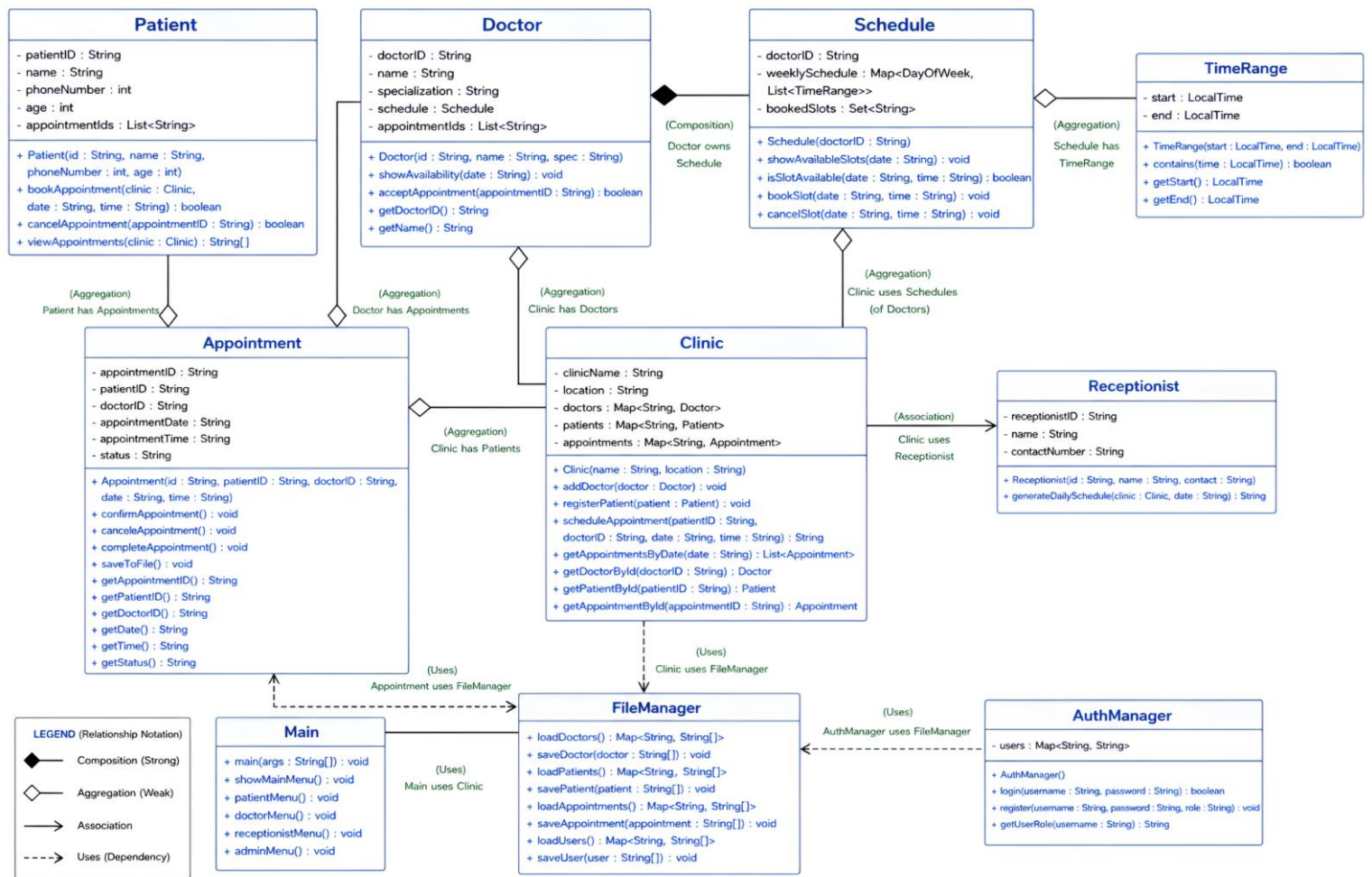
Team Members :

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1. Problem Title : **Clinic Appointment Scheduler System**

STRUCTURAL DESIGN

The system is designed using an object-oriented approach with multiple interacting classes:
Patient, Doctor, Appointment, Receptionist, Schedule, and Clinic.



2. Class diagram (draw class diagram that shows different classes and their relationships. Use UML notations)

BEHAVIOURAL DESIGN

3. Use-case diagram

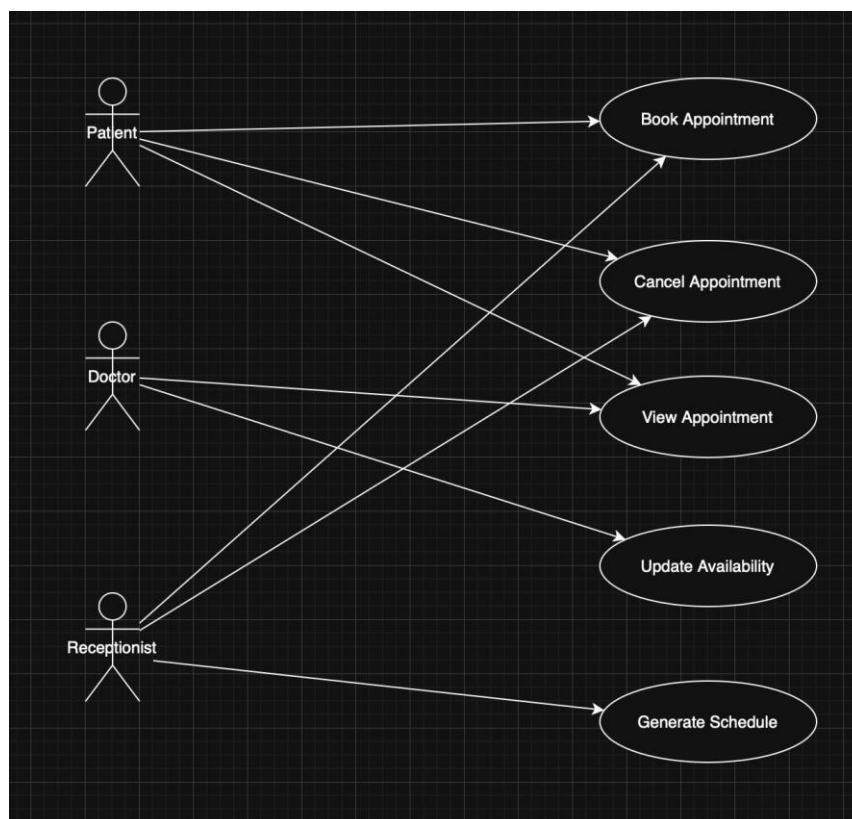
a. List of actors

1. **Patient:** An individual who books, cancels, or views appointments with doctors.
2. **Doctor:** A medical professional who manages availability and accepts or rejects appointments.
3. **Receptionist:** A staff member responsible for managing schedules and generating daily reports.
4. **System Administrator (Clinic Management):** Oversees overall clinic operations including doctor registration and patient management.

b. List of use-cases

1. **Register Patient:** A new patient is added to the clinic system with personal details.
2. **Add Doctor:** The clinic management adds doctors along with their specialization and availability.
3. **Book Appointment:** A patient selects a doctor and schedules an appointment based on available slots.
4. **Cancel Appointment:** A patient cancels an already booked appointment.
5. **View Appointments:** A patient checks their scheduled appointments.
6. **Update Availability:** A doctor updates their available time slots.
7. **Accept Appointment:** A doctor reviews and confirms appointment requests.
8. **Check Availability:** The system checks available slots for a specific doctor.
9. **Generate Daily Schedule:** The receptionist generates a report of all appointments for a given day.
10. **Reschedule Appointment:** An existing appointment is modified to a new date or time.

c. Use case diagram (use UML notations)

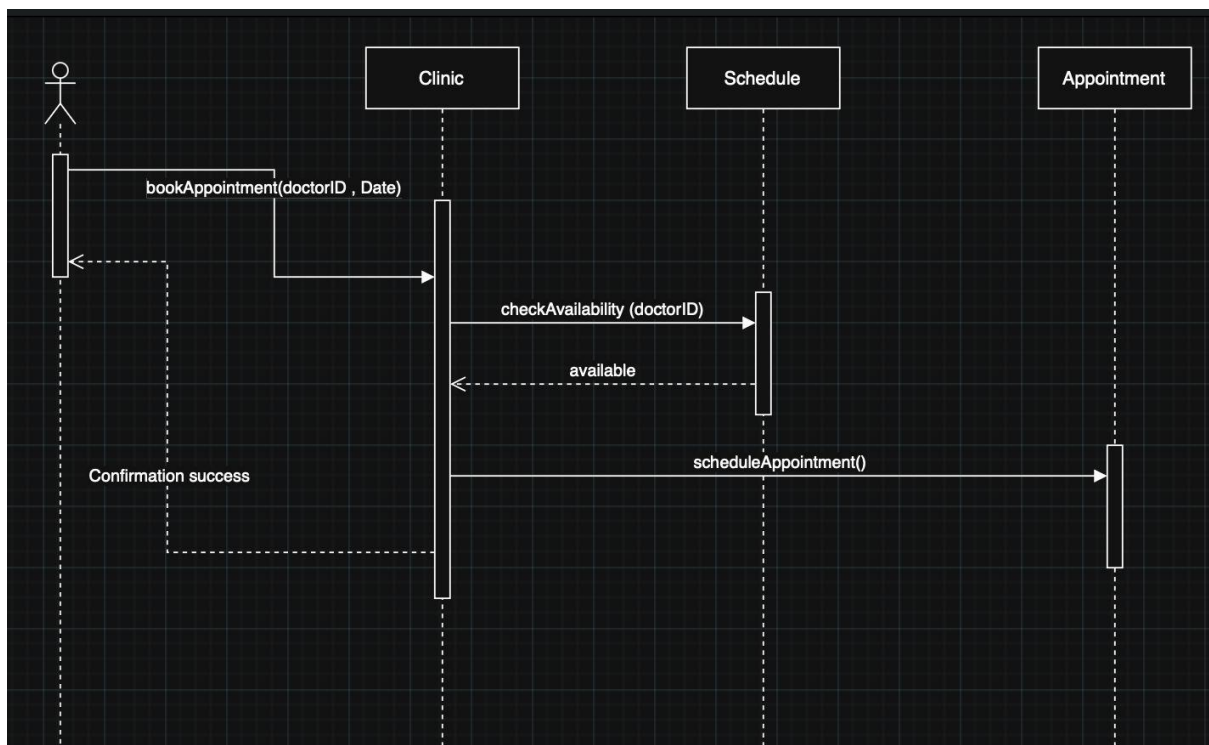


4. Sequence diagram (use UML notations)

Choose key use cases and draw their sequence diagram. First describe the scenario in few sentences. Then draw the sequence diagram.

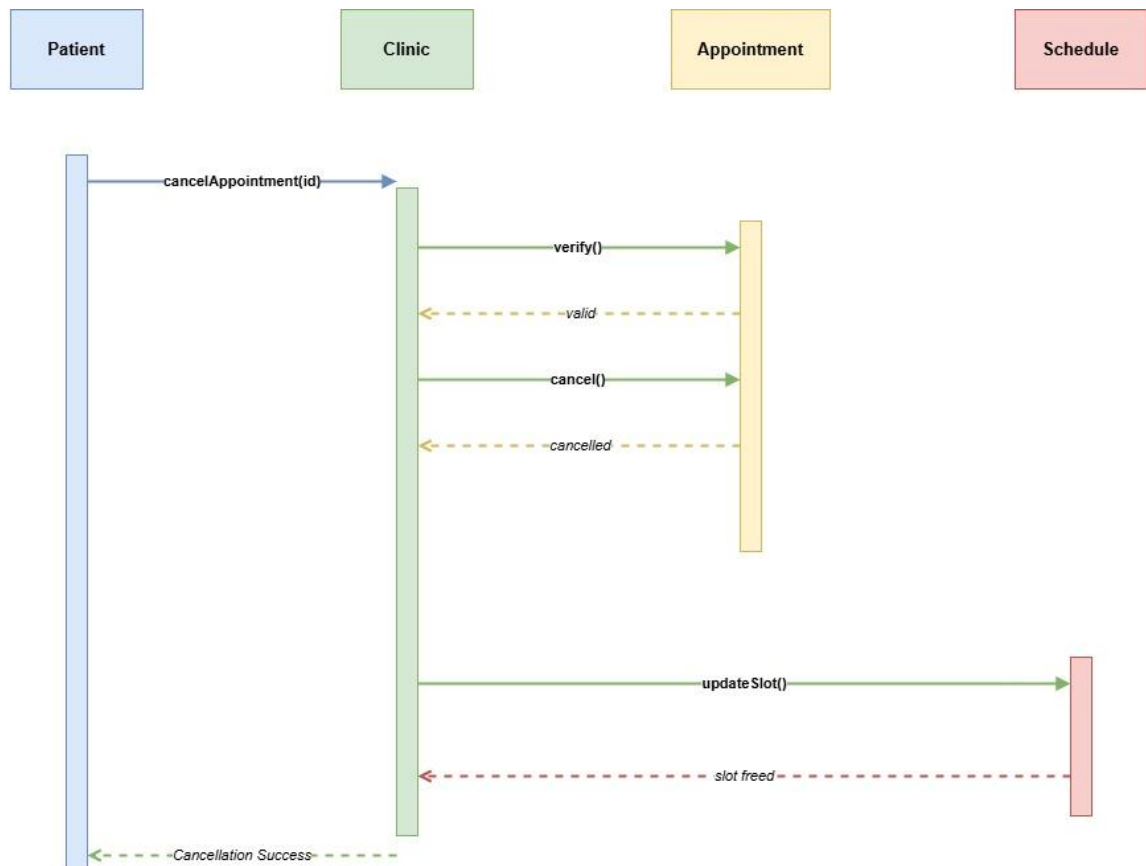
Scenario A: Booking an Appointment

A patient wants to book an appointment with a doctor. The patient sends a request with doctor ID and date to the clinic system. The clinic checks the doctor's availability through the schedule. If the slot is available, the system creates and schedules the appointment, and finally sends a confirmation message back to the patient.



Scenario B: Cancel Appointment

A patient decides to cancel an already booked appointment. The patient sends a cancellation request with the appointment ID. The clinic system verifies the appointment, updates its status, removes the reserved slot from the schedule, and confirms the cancellation to the patient.



Appointment Cancellation — Sequence Diagram

Scenario C: Generate Daily Report

The receptionist wants to view all appointments scheduled for a particular day. The receptionist requests the daily schedule. The system fetches all appointments for that date, compiles them into a report, and displays it.

Daily Schedule Generation — Sequence Diagram

